BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 188 · Binary search: trace and code

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Does linear search require ordered input?

Q2. What must be true of the two input runs when merging them?

Q3. For seven elements, what is the last zero-based index?

Q4. Find the integer midpoint of low=2 and high=8.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

precondition — Input assumption: the array is ordered.

low — Lower inclusive search index.

high — Upper inclusive search index.

mid — Index of the middle candidate.

O(log n) — Search repeatedly halves the candidate set.

New learning

Sort before searching — Binary search relies on a sorted array; otherwise discarding half may lose the target.

Inspect the middle — Compare target to the midpoint and keep only the half that could contain it.

Stop correctly — Return an index when found; return -1 if low becomes greater than high.

Worked example

Search for 19 in [3,7,10,15,19,23,27]. Record every midpoint checked.

1. Predict first — Use inclusive low and high indexes.

2. Model solution — mid 3 → 15, so low=4; mid 5 → 23, so high=4; mid 4 → 19. Return 4.

We Do · Practise with guidance

Search for 9 in [1,4,9,14,18,22,29,33,40].

Predict first

Which midpoint is checked first?

Model solution

mid 4=18, then mid 1=4, then mid 2=9. Return index 2.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Search for 8 in [2,5,8,11,14]. List midpoint indexes.

Question 2

Search for 13 in [2,4,6,8,10,12,14]. List indexes checked and final return.

Question 3

Explain why setting low = mid instead of mid + 1 can cause failure.

Exit Ticket · Answer before checking

Binary search is always safe on an unsorted array.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. No. It scans sequentially and works on unsorted arrays.

Q2. They must already be sorted.

Q3. 6.

Q4. 5: 2 + (8-2)/2 = 5.

You Do pages · Independent questions

Q1. mid=2, match at index 2 on the first comparison.

Q2. mid 3=8, then mid 5=12, then mid 6=14; the window becomes empty, return -1.

Q3. With low=mid and only two candidates left, mid may equal low repeatedly; the interval does not shrink, potentially creating an infinite loop.

Exit Ticket · Model answer

False. It relies on sorted ordering.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: